Structural certificates for the S_{10} atlas table

L054 finite application

Theorem 1. *Relative to the acquired numerical catalogue and structural registry, the* 1593 *rows of the current companion table headed S_{10} have the displayed structural descriptions. An extension expression asserts a normal subgroup, quotient and the stated splitting behaviour. It need not specify a unique abstract isomorphism class.*

Source and acquired results. The companion table by Thomas Connor and Dimitri Leemans is dated 27 November 2012 and accompanies [1]. Its numerical catalogue, source-row bijection, actual embedded generators and subgroup orders are acquired from the separately accepted computation in degree 10. The acquired source contains 1593 conjugacy classes and 8023 weighted maximal-subgroup edges. The subgroup census and incidence counts are not recomputed here; proper-group orders are checked again when constructing literal transports. The registry contains 676 finite structural models reconstructed from accepted earlier applications; the exact files read to reconstruct it are bound to those reviews. Existing hypotheses and qualifications remain part of the acquired statements.

The source article's Section 3.2 gives the convention of left association for an unparenthesized sequence of extensions. This convention is acquired from its separately reviewed native reading. A new audit joins the native powers, subscripts, parentheses and operators of every structural cell to its expression and order. The table has 103 pages and 38011 retained text states. Native pages $1, 6, 33, 52, 102, 103$ were directly inspected. This does not amount to visual inspection of all pages. The position audit uses the same pypdf decoder as the flat extraction, rather than an independent extraction engine. Rows 77 and 576 put redundant parentheses around the single atom D_{10}; only that exact glyph sequence is removed when comparing their expressions. The raw positions are retained. The pinned source edition, rather than every historical edition, specifies the claim.

Lemma 1. *Let H be generated by specified permutations, let S be the union of their supports, and put $n = |S|$. Suppose that the order of this same embedded group is known. If $|H| = n!$, then $H = \mathrm{Sym}(S)$. If every specified generator is even and $|H| = n!/2$, then $H = \mathrm{Alt}(S)$.*

Proof. Every generator fixes the complement of S, so H is a subgroup of $\mathrm{Sym}(S)$. Equality of finite orders proves the first claim. The sign map is a homomorphism. If all generators are even, then $H \leq \mathrm{Alt}(S)$, and equality of orders again gives equality of groups. No transitivity assumption is needed. In the application the source row is first joined to its accepted numerical record and exact generator tuple. The acquired order is then the order of the group in this lemma; it is not inferred from the printed name. □

This identifies rows $1, 2, 3, 4, 7, 11, 29, 45, 109, 191, 451, 738, 1397, 1547$ directly. The other 1579 source occurrences are handled using the acquired models, 172 new finite model certificates and complete permutation transports. The model roster consists of 87 split extensions, 69 direct products and 16 nonsplit extensions. In particular, an abstractly symmetric or alternating subgroup with another action is not accepted by the support lemma unless its actual support and order meet the stated conditions.

Lemma 2. *Let M and H be finite groups given by complete closures of specified permutation generators. A bijection $f : M \to H$ satisfying $f(1) = 1$ and $f(xg) = f(x)f(g)$ for every $x \in M$ and every specified generator g is an isomorphism.*

Proof. Every element is a positive word in the generators, because the inverse of a generator in a finite group is a nonnegative power of it. Induction on such a word gives $f(xy) = f(x)f(y)$. The bijection is therefore an isomorphism. The reader constructs proposed images along the complete closure, checks consistency whenever a word reaches an old element, and checks every right-generator edge. It compares the entire image with the actual target subgroup. Merely mapping a generating set is insufficient. □

Products and extensions. For a new model H, the reader closes the displayed generators of N, checks $N \triangleleft H$, and transports an earlier proved model onto N. A split certificate also supplies K, its structure, $N \cap K = 1$ and $|N||K| = |H|$. Normality makes NK a subgroup; its order then gives $H = NK$. For a colon, the retained convention requires nontrivial conjugation of K on N, witnessed by a pair of noncommuting generators. A direct product requires every pair of factor generators to commute, which implies that the factors commute elementwise. The finite transport and extension predicates are reused from the accepted common method. Every newly proposed model is proved before it is used, including models obtained in an earlier candidate search.

Twelve nonsplit models have a prime quotient of order p. A complement exists exactly when there is an element of order p outside N: its cyclic subgroup intersects N trivially, and the order formula gives all of H. The nonsplitting reader checks every outside element, so the absence of such an element proves the assertion.

The other four nonsplit models have quotient S_3. Their specified quotient generators satisfy $a^2 = b^3 = (ab)^2 = 1$ and generate the quotient. These are the standard defining relations of S_3. A complement would lift them to elements satisfying the same relations. Conversely such lifts give a homomorphism from S_3 whose composite with the quotient map is the identity, hence a section. All lifts are considered; those failing the first two relations may be discarded before testing the third. The recorded count includes the full original Cartesian product. Each possible pair fails at least one relation, so no section exists.

The proposal search. The first pass stopped at row 34, with expression $A_5 \times A_5$: GAP's general complement routine does not handle the case where both the normal factor and the quotient are nonsolvable. A process exit code of zero after the resulting break-loop error is recorded as an incomplete search. The retained witnesses are not discarded or accepted on that basis.

The following elementary criterion supplies a suitable direct-product search. Let $N \triangleleft H$ and $n = [H : N]$. If $C_H(N)$ has order n and intersects N trivially, it is the unique direct complement of N. Indeed, a direct complement lies in that centralizer and has the same order. Otherwise one can inspect the normal subgroups K of order n with $N \cap K = 1$. Normality puts every commutator in the intersection, so these factors commute, and the order formula gives $H = NK$. Conversely every internal direct complement is normal, since conjugation by N fixes it pointwise and conjugation by itself preserves it. This standard direct-product criterion changes proposal generation only. The literal proof predicates remain unchanged.

Verification and attribution. The numerical catalogue, source expressions and association convention belong to the source and acquired results. The support criterion and the internal direct-product criterion above are elementary group theory. The contribution of this application is an explicit source join and finite certificates for its remaining occurrences using the common method. The complete literal check passed in 22.115 seconds, with 1593 source joins, 14 support identifications, 172 model proofs, 1907 transports and 3209070 generator edges. Nonsplitting requires 10236 outside-element tests and 5764 quotient-lift pairs. Inputs remained unchanged and stderr was empty. The initial incomplete candidate run is retained; the 27 remaining rows

were subsequently supplied in 2.531 seconds using the direct-factor improvement. Candidate generation itself is never proof.

Ten damaged source inputs are rejected, and eight narrow glyph checks confirm that only the two parenthesized D_{10} atoms are normalized. Support recognition has its separate eight small literal model controls and nine damaged cases. Ordinary and optimized modes are checked separately. The final independent review records source admission and retains the extraction limitations stated above. No conjecture resolution, Lean formalization, historical-byte identity, whole-article completion or compression measurement is claimed. Other large companion tables remain in scope for L054.

References

[1] T. Connor and D. Leemans, An atlas of subgroup lattices of finite almost simple groups, *Ars Mathematica Contemporanea* **8** (2015), 259–266. The retained companion table is headed S_{10} and dated 27 November 2012.